HakaseAI | BIO

PRE-CLINICAL BIODIGITAL TWIN · PRODUCT DOCUMENTATION

# The AI co-scientist for pre-clinical discovery.

Hakase Bio simulates the journey from target to organism so your team can screen, rank and de-risk candidates in silico — then confirm the best of them at the bench. It is built to **sharpen the judgment of your scientists and CROs, not to replace them.**

CELLULAR SIMULATION | STRUCTURAL PREDICTION | ORGANISM PBPK | ADMET & SAFETY | PREDICT-THEN-CONFIRM

**Hakase Bio** · A decision-support layer for Discovery → IND.
Every prediction ships with the experiment that would test it.

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For client & partner distribution

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## 01 What Hakase Bio is

A decision-support layer for the pre-clinical team

Hakase Bio is a pre-clinical BioDigital Twin — an AI/ML platform that simulates how a drug candidate behaves from the cell, through its target structure, to the whole organism. It gives discovery teams an early, quantitative read on which hypotheses are worth a wet-lab experiment, and which are likely dead ends, before resources are committed.

Think of it as a co-scientist that never tires of reading the literature, modelling the biology and running the numbers. It proposes where to look first, explains why, and — crucially — names the experiment that would prove it wrong. Your scientists stay in command of every decision; Hakase Bio simply makes each decision better informed and faster to reach.

| 10× | –65% | 92% |
|---|---|---|
| **Hypotheses screened** | **Wet-lab cycles to a lead** | **Cross-assay accuracy** |
| Per scientist, per week‡ | Fewer dead-end experiments‡ | vs. held-out CRO readouts‡ |

### What it is — and what it is not

**Hakase Bio is…**

- \+ A **decision-support and prioritisation** layer for discovery through IND.
- \+ A way to **spend wet-lab time on the best ideas** instead of screening the field blind.
- \+ An **early-warning system** for toxicity, PK and developability liabilities.
- \+ A **hypothesis engine** that always pairs a prediction with its falsifying experiment.

**Hakase Bio is not…**

- – A **replacement** for your scientists, your bench, or your CROs.
- – An **autonomous decision-maker** — humans approve every call.
- – A **substitute for wet-lab or animal confirmation** of a candidate.
- – A **black box** — every prediction is explained and traceable to evidence.

## 02 The pre-clinical problem

Why so much bench effort is spent on candidates that were never going to make it

Bringing a single medicine to patients now costs on the order of **$1–2.6 billion and takes 10–15 years.**[1,2] The dominant reason is not a shortage of ideas — it is attrition. Roughly **nine in ten candidates that enter clinical trials never reach patients,** and only about one in ten compounds leaving pre-clinical development is ultimately approved.[3,4]

| $1–2.6B | ~90% | ~75% | 10–15yr |
|---|---|---|---|
| Cost per approved drug[1,2] | Clinical candidates that fail[3] | Failures from efficacy or toxicity[4] | From target to market[1] |

When candidates fail, the causes are remarkably consistent: roughly **40–50% for lack of efficacy, ~30% for safety or toxicity, and ~10–15% for poor pharmacokinetics.**[3,4] Most of these liabilities are, in principle, detectable early — yet they routinely surface in the costly late stages, where a single Phase III failure can cost **$350M to $1B+.**[2] The earlier a doomed candidate is identified, the more capital, time and patient exposure is saved.

### The wet lab is precious — and finite

Every assay, every animal study and every CRO engagement consumes scarce time, budget and ethical capital. The bottleneck is rarely the experiment itself; it is deciding *which* experiments to run. Teams are forced to triage hundreds of plausible hypotheses with limited capacity, and good ideas are routinely starved while resources are spent confirming candidates that a deeper look would have de-prioritised.

> **THE OPPORTUNITY**
>
> If a team could reliably rank candidates and surface their fatal flaws *before* plating a well, the same bench capacity would produce far more validated leads. That is precisely the gap Hakase Bio is built to close — by making the in-silico read trustworthy enough to guide where the bench goes next.

## 03 Our philosophy — augment, don't replace

The scientist stays in command; the model does the legwork

The industry's most durable insight about AI in biology is that prediction and experiment are partners, not competitors. The emerging standard — often called **"lab-in-the-loop"** or **human-in-the-loop** discovery — pairs computational speed and pattern recognition with the judgment of experienced scientists and the ground truth of the bench.[5,6] AI proposes and ranks; humans decide; the wet lab confirms; and every result sharpens the next prediction.

Hakase Bio is built natively around this loop. It does not try to be an oracle that ends experimentation. It is a **co-scientist** that compresses the search space so your team's finite bench capacity is aimed at the candidates most likely to succeed — and so that when an experiment runs, it is the one most capable of settling the question.

### Three commitments that keep the human in charge

- **Falsifiable by default**

  If a prediction cannot be tested, it does not ship. Every output carries the experiment that would prove it wrong, so the bench always has a clear next move.

- **Explainable, not opaque**

  Each prediction is traced to the data, structure and pathway evidence behind it. Scientists can interrogate the "why," not just the score.

- **Confirmation is ground truth**

  Wet-lab, CRO and animal readouts are treated as the source of truth that closes the loop — never overruled by the model.

> **IN ONE LINE**
>
> Hakase Bio tells your team where to look first and what would prove it wrong — so the bench spends its time confirming the best ideas, not discovering the dead ends.

## 04 How it works — the target-to-organism cascade

Three modelling layers, each gated by the biology beneath it

Hakase Bio is organised as a connected cascade. A candidate is evaluated first at the level of the cell, then its molecular structure, then the whole organism. A prediction only advances to the next layer where the biology supports it — so weak hypotheses are filtered early, and the candidates that reach the top of the funnel are supported by evidence at every level.

| | |
|---|---|
| L1 CELLULAR | **Cellular AI simulation**<br>Single-cell foundation models — the same class of approach behind today's "virtual cell" research[7] — with integrated reference atlases and pathway-aware priors predict how cells respond to a perturbation, before any cells are cultured. |
| L2 STRUCTURAL | **Structure × tissue gate**<br>Predicted protein–ligand structures and designed binders are gated by tissue-level expression and proteomics, so a candidate only progresses where the target is actually present and the interaction is structurally credible. |
| L3 ORGANISM | **Whole-body PBPK**<br>Physiologically based pharmacokinetic (PBPK) and IVIVE engines translate in-vitro signals into whole-body behaviour — projecting human pharmacokinetics, first-in-human dose priors and cross-species bridging,[8] the same modelling class regulators increasingly accept. |

Because the layers share one representation of the candidate, a finding never has to be re-keyed or re-interpreted as it moves up the cascade. The result is a single, coherent profile — from molecular plausibility to predicted human dose — that a scientist can read end to end.

## 05 Safety & developability profiling

Surfacing the liabilities that sink late-stage programs — early

Because toxicity and pharmacokinetics together account for a large share of attrition, Hakase Bio profiles every candidate for safety and developability up front. The goal is simple: give teams the chance to catch a fatal flaw at the in-silico stage, when changing course is cheap, rather than in the clinic, when it is not.[4,9]

### Structural safety alerts

Screens for mutagenicity (Ames) and structural liabilities associated with liver injury (DILI) and cardiac (hERG) risk.

### Machine-learning ADMET

An ML engine predicts dozens of absorption, distribution, metabolism, excretion and toxicity endpoints directly from structure.

### Developability read

Physicochemical and developability properties flag candidates that will be difficult to formulate or progress.

### Early IND-readiness signal

A consolidated safety profile gives programs an early, defensible read on regulatory readiness.

Each flag is a prompt for the bench, not a verdict. A predicted hERG liability becomes a targeted assay to confirm or refute; a favourable ADMET profile becomes confidence to advance. The model narrows the questions — the laboratory answers them.

## 06 Predict-then-confirm

The defining workflow — and the reason the bench stays in command

What separates Hakase Bio from a conventional "score and serve" model is what happens after a prediction. Most platforms rank a list and stop; the team picks one, runs the experiment, and the model is never told whether it was right. Hakase Bio closes that loop deliberately.

**01 · PREDICT**

**Simulate & rank**

The cascade screens hypotheses and ranks candidates with calibrated confidence.

**02 · FALSIFY**

**Name the test**

Each prediction ships with the specific experiment that would prove it wrong.

**03 · CONFIRM**

**Run at the bench**

Your scientists or CRO run that experiment. The readout is the ground truth.

**04 · UPDATE**

**Close the loop**

The result feeds back; the twin refines, and the next prediction is sharper.

▸ PREDICT → FALSIFY → CONFIRM → UPDATE → REPEAT ◂

This design has three consequences that matter to a research leader. First, **the bench is always in control** — the model proposes the test, but the experiment decides. Second, **quality compounds**: every confirmation makes the engine more accurate on your chemistry and biology, rather than leaving it static. Third, **nothing is unfalsifiable** — if a claim cannot be tested, it is not shipped, which keeps the platform honest and auditable.

> **WHY IT BUILDS TRUST**
>
> A prediction you can immediately test is a prediction you can trust. By pairing every output with its falsifier, Hakase Bio invites the scrutiny of the bench instead of asking teams to take a score on faith.

## 07 Why Hakase Bio leads

Where it sits in the category — and where it pulls ahead

The AI-for-discovery field — a market growing at **25–30%+ a year**[10] — has produced excellent specialists: physics-based chemistry platforms, phenomics-first screening, and structure-prediction engines. Each is strong at one layer of the problem. Hakase Bio's advantage is architectural: it is built as a **connected cascade across all three layers**, joined by a confirmation loop, rather than a point tool for a single stage.

| CAPABILITY | TYPICAL POINT TOOLS | HAKASE BIO |
|---|---|---|
| **Scope of modelling** | One layer — chemistry, phenomics, or structure | **Full cell → structure → organism cascade** |
| **Relationship to the bench** | Score-and-serve; loop rarely closed | **Predict-then-confirm with a built-in falsifier** |
| **Explainability** | Often a ranked list with limited rationale | **Every prediction traced to its evidence** |
| **Safety & PK** | Frequently a separate, bolt-on step | **ADMET, tox and PBPK native to the cascade** |
| **Regulatory posture** | Research output, not submission-shaped | **Aligned to MIDD & credibility frameworks** |
| **Data & IP** | Varies; data sometimes pooled | **Tenant-isolated; never trains on your data** |

Comparison describes general patterns across the category, not any single named competitor. Capabilities vary by vendor and evolve quickly.

## 08 Validation & scientific credibility

Built to be checked, not just believed

A model is only as useful as it is trustworthy, and trust in this domain is earned through evidence. Hakase Bio is designed to be validated continuously and held to recognised standards rather than asked for the benefit of the doubt.

### Held-out benchmarking

Predictions are assessed against held-out experimental readouts and external reference datasets, with calibrated confidence so teams know when to trust a call — and when to test it.

### Grounded in established science

The cascade builds on peer-reviewed method classes — single-cell foundation models, structure prediction and PBPK/IVIVE — not bespoke heuristics.

### Credibility-framework aligned

Designed to support model-credibility expectations such as ASME V&V40 and MID3, so evidence can travel toward a submission.

### Regulatory tailwind

Regulators increasingly accept model-informed approaches and PBPK in review,[8,11] and the field is converging on hybrid mechanistic-plus-AI methods.

**AN HONEST POSTURE ON ACCURACY**

In-silico models inform decisions; they do not end them. Hakase Bio's role is to raise the prior probability of success and concentrate experimental effort — with the bench retained as the final arbiter for any candidate that advances.

## 09 Where it fits in your workflow

From target triage to IND-enabling decisions

Hakase Bio slots into the discovery pipeline you already run, adding a decision-support layer at the points where bench capacity is allocated.

### Target triage

Rank and de-risk targets by cellular response and tissue relevance before committing a program.

### Hit-to-lead

Prioritise the chemistries most likely to confirm, and surface liabilities before they cost a cycle.

### Lead optimisation

Balance potency against ADMET and safety, with each trade-off paired to a confirming assay.

### Candidate selection

Compare finalists on a single end-to-end profile, from structure to predicted human dose.

### First-in-human dose

PBPK/IVIVE priors inform FIH dose discussions and species bridging.

### IND readiness

Assemble an early, defensible safety and developability picture as a program approaches IND.

## 10 Outcomes & who it's for

The teams that benefit, and what they gain

**10×**
**More hypotheses explored**
Same bench capacity‡

**–65%**
**Fewer wet-lab cycles to a lead**
Dead ends caught earlier‡

**Earlier**
**Liabilities surfaced**
When course-correction is cheap

### Discovery biologists

Explore more of the hypothesis space and focus assays where they will move a program.

### Med-chem & DMPK

See potency, ADMET and safety trade-offs together, each tied to a confirming experiment.

### Translational & pre-clinical

Carry one coherent profile from cell to predicted human dose into IND-enabling decisions.

### R&D leadership

Allocate finite bench and budget toward the candidates most likely to succeed.

### Biotech founders

Stretch scarce capital further by de-risking candidates before expensive studies.

### CRO partners

Receive sharper, pre-prioritised experimental plans — not more noise.

## 11 Security, data & IP

Your chemistry and biology stay yours

Pre-clinical data is among the most valuable IP a team holds. Hakase Bio is built so that using the platform never means surrendering it.

### Tenant-isolated by default

Each customer's environment is isolated. Data is not pooled across tenants.

### Never trains on your data

Your candidates and results are used to serve you — never absorbed into shared model weights.

### Confirmations as ground truth

Experimental readouts close your loops and improve your results, not a shared model.

### Traceable & auditable

Every prediction-to-confirmation pairing is versioned and exportable for internal review.

**Tenant isolation** · **No-train-on-customer-data** · **V&V40 credibility** · **MID3 evidence** · **Audit-ready provenance**

Specific certifications and data-handling terms for a given deployment are confirmed during onboarding.

## 12 Get started

A working session on a problem you care about

The best way to evaluate Hakase Bio is to watch it run a real predict-then-confirm loop on a target or candidate from your own pipeline. We will set up a working session, walk through the cascade end to end, and show you the falsifier it proposes — so you can judge the science against your bench, not a slide.

BRING HAKASE BIO TO YOUR NEXT PROGRAM

### Sharpen the bench. Keep the scientist in command.

Whether you are triaging targets, optimising leads, or preparing for IND, Hakase Bio helps your team aim its finite experimental capacity at the candidates most likely to succeed — and tells you exactly how to test each call.

REQUEST A DEMO

**A guided walkthrough on your target**

PARTNER WITH US

**Co-develop on a live discovery program**

PLATFORM

**Hakase Bio · the pre-clinical BioDigital Twin**

‡ Figures marked are indicative platform benchmarks intended to illustrate typical impact; they are not a guarantee of results for any particular program. Scope, validation and outcomes are defined per engagement. This document is intended for client and partner distribution.

APPENDIX

# Selected references & industry sources

Industry figures cited in this document are drawn from the public sources below. They describe the broader drug-development landscape and are provided for context.


1 **Congressional Budget Office.** Research and Development in the Pharmaceutical Industry. cbo.gov/publication/57126.

2 **RAND Corporation (2025); DiMasi et al.** Cost of drug development and per-phase trial cost estimates. rand.org; pubmed.ncbi.nlm.nih.gov/32125404.

3 **Sun D. et al. / Pharma Excipients.** Why ~90% of clinical drug development fails. pharmaexcipients.com; asbmb.org.

4 **Attrition root-cause analyses.** Efficacy (~40–50%), safety/toxicity (~30%), pharmacokinetics (~10–15%). Wiley Online Library; PMC7517894.

5 **Inductive Bio; Ginkgo Bioworks (2025).** AI-driven "lab-in-the-loop" discovery workflows. inductive.bio.

6 **Human-in-the-loop AI discovery.** Collaborative AI + expert-chemist validation; FutureHouse "Robin." nvidia.com; drainpipe.io.

7 **Single-cell foundation models.** scGPT (Nature Methods, 2024); Geneformer; virtual-cell research. czi virtualcellmodels.

8 **PBPK & first-in-human dose modelling.** PBPK for FIH dose selection and species bridging. PMC8957247; accp1 (CPDD, 2025).

9 **Computational toxicity prediction.** Early ADMET/tox to reduce late-stage attrition. PMC11187637; pubs.acs.org.

10 **AI-in-drug-discovery market.** ~$2–6B (2025), ~25–30%+ CAGR. Grand View Research; Precedence Research; BioSpace.

11 **Model-informed drug development (MIDD).** Growing FDA/ICH acceptance; ICH M12 (2024); ASME V&V40 credibility. PMC12655628; accp1 (CPDD, 2025).